

TAO LIN

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EDUCATION

Zhejiang University

B.Eng. in Computer Science and Technology

Oct. 2012 - Present

Hangzhou, China

Overall GPA: 3.96 / 4.00

The third year GPA: 4.00 / 4.00

Rank: 1st in Qizhen Honors Class, Chu Kochen Honors College (Top 2%)

Hong Kong University of Science and Technology

Exchange Student for Final Year Project

Sept. 2015 - Feb. 2016

Hong Kong

EXPERIENCE

Visual Analytics Group, State Key Lab of CAD&CG

Research Assistant

June 2013 - Aug. 2015

Zhejiang University, China

Research Advisor: Professor Wei Chen [\[Homepage\]](#)

- Administrated a computer cluster with 20 nodes for one-plus year.
- Processed large-scale data on the cluster with Hadoop and Spark for one-plus year.
- **Project Mobility Viewer** - A visual analytics system supporting situation-aware understanding and visual reasoning of human mobility. [\[PDF\]](#) [\[Video\]](#) *Core Member*
 - Preprocessed tera-scale data on the cluster by Spark.
 - Designed a distributed algorithm to detect and eliminate ping-pong effects.
 - Designed and implemented the Flow Volume Analysis Subsystem.
 - Technique: MapReduce, Hadoop, Spark, Python, JavaScript (D3, WebGL), OpenStreetMap.
- **Project TieVis** - A visual analytics framework and system that enables interactive analysis and exploration of the dynamic changes of interpersonal ties. [\[PDF\]](#) [\[Video\]](#) [\[Code\]](#) *Project Leader*
 - Preprocessed the data by Python (Extract, Aggregate, PCA and MDS).
 - Designed and implemented the whole TieVis System.
 - Technique: Python(NumPy, SciPy, scikit-learn), JavaScript.

Multimedia Technology Research Center

Visiting Intern

Sept. 2015 - Feb. 2016

HKUST, Hong Kong

Supervisor: Professor S.-H. Gary Chan [\[Homepage\]](#)

- Worked for Streamphony, a next-generation streaming cloud for large-scale high bit-rate applications over the global Internet.
- Developed the research results into a practical system for commercial trials.
- Implemented the Forward Error Correction (FEC) Module by Reed-Solomon Code.
- Improved the quality and robustness of the video stream.
- Technique: C++ (STL, Boost (asynchronous, memory pool and smart pointer)), Lock-Free Programming, Code Review, Unit Test(Google Test, Google Mock).

PUBLICATION

Mobility Viewer: A Eulerian Approach for Studying Urban Crowd Flow

Yuxin Ma, **Tao Lin**, Zhendong Cao, Chen Li, Wei Chen

IEEE Transactions on Intelligent Transportation Systems. Accept, 2015. [\[PDF\]](#) [\[Video\]](#)

TieVis: Visual Analytics of Evolution of Interpersonal Ties

Tao Lin, Fangzhou Guo, Yingcai Wu, Wei Chen, Biao Zhu, Sidong Wang, Huamin Qu

The 9th IEEE Pacific Visualization Symposium (PacificVis 2016). Submitted. [\[PDF\]](#) [\[Video\]](#) [\[Code\]](#)

HONORS AND AWARDS

Google Excellence Scholarship <i>58 Bachelor and Master students in mainland China</i>	2015 <i>Google Inc.</i>
National Scholarship <i>Top 1% students</i>	2014 - 2015 <i>Ministry of Education, China</i>
First-Class Scholarship for Outstanding Students <i>Top 3% students</i>	2013 - 2014, 2014 - 2015 <i>Zhejiang University</i>
Honorable Mention, Interdisciplinary Contest in Modeling	2014
Second Prize, Zhejiang University Programming Contest	2014
Third Prize, ACM-ICPC China Zhejiang Provincial Programming Contest	2014

SELECTED PROJECTS

MiniSQL Database Management System <i>Team Leader</i>	Sept. 2014 - Nov. 2014 <i>Course Project</i>
· Designed and implemented a DBMS, which can create/delete/query tables/indexes/records by interpreting SQL. · Modules: Interpreter, API, Catalog Manager, Record Manager, Index Manager, Buffer Manager. · Technique: C++, Parser, Cache Replacement (LRU), B+ Tree.	
Visual Analysis of Library Data <i>Team Leader</i> [Video] [Code]	April 2014 - May 2015 <i>Student Research Training Program</i>
· Promoted library users' decision-making by interacting with real records. · Designed and implemented a visual analysis system. · Technique: JavaScript (D3, Bootstrap, ECharts and jQuery), Python (Django), Oracle Database.	
Precomputed Shadow Fields for Dynamic Scenes <i>Independent</i> [Code]	Mar. 2015 - July 2015 <i>Course Project</i>
· Implemented a system for rendering dynamic soft shadows on the basis of a paper in SIGGRAPH 2005. · Course: Advances in Computer Graphics, Score: 94/100, Instructor: Kun Zhou. · Technique: C++ (DirectX), HLSL, Precomputed Radius Transfer, Spheric Harmonics Function.	
A 3-D Auto-Stereoscopic Front Projection System <i>Team Member</i> [Video]	July 2014 - May 2015 <i>Challenge Cup Competition</i>
· Achieved 3-D views with unaided eyes by 60 projectors, with teammates majoring in Optical Engineering. · Derived the core mathematical formulas by light field reconstruction principles. · Implemented the prototype by OpenGL.	

TECHNICAL STRENGTHS

Computer Languages	C/C++, JavaScript, Python, Java, MATLAB, Verilog
Libraries & Frameworks	Visualization (D3, ECharts) C++ Unit Test (Google Test, Google Mock)
Tools	Computer Graphics & Vision (OpenGL, DirectX, WebGL, OpenCV) Data Science (Hadoop, Spark, Hive, MySQL) Software Engineering (Unix Command, Git, Phabricator)